Algebraic Language for Representation of Quantum Circuits

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1 Preliminaries

The minimal information unit within a quantum computing system is the qubit, which it is denoted by a unitary vector within a Hilbert space of dimension 2 denoted by $\mathcal{Q}$. Multiple qubits can be taken together so that an element of the Hilbert space of n qubits is considered and denoted by $\mathcal{Q}_n$. This case can be seen as the tensor product of the n Hilbert spaces, $\mathcal{Q}_n = \mathcal{Q}^{\otimes n} = \mathcal{Q} \otimes \cdots \otimes \mathcal{Q}$ corresponding to each qubit in the system, so that the resulting space dimension is 2^n.

Let $\mathbb{N}$, $\mathbb{R}$ and $\mathbb{C}$ denote the sets of natural, real and complex numbers, respectively. Since the 0 is not a natural number, let $\mathbb{N}^*$ be the natural number within cero. The notation $\mathbb{N}_n$ and $\mathbb{N}_n^*$ is used to denote the subset of $\mathbb{N}$ and $\mathbb{N}^*$ respectively that contains the natural numbers smaller or equal than n. In set theory, for any finite set S, its number of elements $|S|$ is called the cardinality of S.

Let Θ denote the set $\{0, 1\}$ and the Kronecker delta is the function $\delta : \mathbb{N}^* \times \mathbb{N}^* \longrightarrow \Theta$ defined by $\delta(k, l) = \delta_{k,l} = 1$ if $k = l$ and 0 otherwise. For a set $\Gamma \subseteq \mathbb{N}_n^*$ the value of $\delta_{k,\Gamma}$ is 1 if $k \in \Gamma$ and 0 otherwise.

2 Basic notation

2.1 Sets

Let $\Gamma = \{k_1, \ldots, k_l\} \subseteq \mathbb{N}_n$ be a set with cardinality $|\Gamma| = l$ and $\gamma = (\gamma_k)_{k=1}^l \in \Theta^l$, llamaremos $\gamma^\Gamma = (\gamma_k^\Gamma)_{k=1}^n$ al elemento de Θ^n definido por:

$$\gamma_k^\Gamma = \begin{cases} \gamma_k & \text{si } k \in \Gamma \\ 0 & \text{si } k \notin \Gamma \end{cases}$$

Por otro lado, si $\gamma \in \Theta^n$, llamaremos γ_Γ al elemento de Θ^l definido por $\gamma_\Gamma = (\gamma_{k_m})_{m=1}^{l}$.

Observemos que para todo elemento $\gamma \in \Theta^l$ se cumple que ${\gamma^\Gamma}_\Gamma = \gamma$ mientras que en general si $\gamma \in \Theta^n$ entonces ${\gamma_\Gamma}^\Gamma \neq \gamma$.

También vamos a considerar que los elementos $(0,\ldots,0) \in \Theta^l$ y $(1,\ldots,1) \in \Theta^l$ lo denotaremos simplemente por 0 y 1.

2.2 Qubits

As introduced in [2], qubits are normally written in the so-called bra–ket notation. Each qubit is represented as a ket, of the form $|a\rangle$, which denotes a vector in $\mathcal{Q}$.

For instance, the computational basis states are written as $|0\rangle$ and $|1\rangle$, and called ket 0 and ket 1, respectively (see [3]). Thus, any qubit can be described by a linear combination of these two basis states as $|a\rangle = \alpha_0 |0\rangle + \alpha_1 |1\rangle$, where $\alpha_0, \alpha_1 \in \mathbb{C}$ such that $|\alpha_0|^2 + |\alpha_1|^2 = 1$.

Notation 2.1. For any qubit $|a\rangle$, the following notation is used:

$$a_0 = \langle 0|a\rangle \text{ and } a_1 = \langle 1|a\rangle,$$

so the qubit $|a\rangle$ can be write by:

$$|a\rangle = a_0 |0\rangle + a_1 |1\rangle \tag{1}$$

For other basis, like for example $\{|+\rangle, |-\rangle\}$, where $|+\rangle = \frac{|0\rangle+|1\rangle}{\sqrt{2}}$ and $|-\rangle = \frac{|0\rangle-|1\rangle}{\sqrt{2}}$, then a qubit $|a\rangle$ can be described like:

$$|a\rangle = a_+ |+\rangle + a_- |-\rangle$$

2.3 kets

In general, for n qubits $|a_1\rangle \ldots |a_n\rangle$, the composite ket $|a\rangle = |a_1 \cdots a_n\rangle \in \mathcal{Q}_n$ can be written as shown below:

$$|a\rangle = a_{k0} |a_1 \cdots a_{k-1} 0 a_{k+1} \cdots a_n\rangle + a_{k1} |a_1 \cdots a_{k-1} 1 a_{k+1} \cdots a_n\rangle. \tag{2}$$

For example, for two qubits $|a\rangle$ and $|b\rangle$, the following can be written:

$$\begin{aligned} |ab\rangle &= a_0 |0b\rangle + a_1 |1b\rangle && (3)\\ &= b_0 |a0\rangle + b_1 |a1\rangle && (4)\\ &= a_0 b_0 |00\rangle + a_0 b_1 |01\rangle + a_1 b_0 |10\rangle + a_1 b_1 |11\rangle && (5) \end{aligned}$$

Notation 2.2. For any ket $|a\rangle = |a_1 \dots a_n\rangle$ of n qubits, the following notation is used:

$$|a\rangle_\gamma^k = |a_1 \cdots a_{k-1} \gamma a_{k+1} \cdots a_n\rangle\,, \tag{6}$$

where $k \in \mathbb{N}_n$ and $\gamma \in \Theta$.

Remark 2.3. The same notation is used, more generally, when γ is an arbitrary single-qubit state: for any $|\gamma\rangle \in \mathcal{Q}$, the ket $|a\rangle_\gamma^k$ denotes the state obtained from $|a\rangle$ by replacing its k-th qubit with $|\gamma\rangle$. For instance, $|a\rangle_+^k$ and $|a\rangle_-^k$ denote the substitution of the k-th qubit by $|+\rangle$ and $|-\rangle$, respectively.

Consequently, the expression (2) can be written as follows:

$$|a\rangle = a_{k0}\,|a\rangle_0^k + a_{k1}\,|a\rangle_1^k\,. \tag{7}$$

For instance, if $|a_1\rangle\,, |a_2\rangle$ and $|a_3\rangle$ are three qubits, then:

- $|a_1\rangle_0^1 = |0\rangle$ and $|a_1\rangle_1^1 = |1\rangle$.
- $|a_1 a_2\rangle_0^2 = |a_1 0\rangle$.
- $|a_1 a_2 a_3\rangle = a_{20}\,|a_1 0 a_3\rangle + a_{21}\,|a_1 1 a_3\rangle = a_{20}\,|a\rangle_0^2 + a_{21}\,|a\rangle_1^2$.
- $|00 a_3\rangle_1^3 = |001\rangle$.

Notation 2.4. Para una notación más general basado en conjuntos de índices, dado el conjunto $\Gamma = \{k_1, \dots, k_l\}$ y $\gamma \in \Theta^l$ denotamos por:

$$|a\rangle_\gamma^\Gamma = ((1-\delta_{1\Gamma})\,|a_1\rangle + \delta_{1\Gamma}\,|\gamma_1^\Gamma\rangle) \otimes \cdots \otimes ((1-\delta_{n\Gamma})\,|a_n\rangle + \delta_{n\Gamma}\,|\gamma_n^\Gamma\rangle)$$

Veamos exactamente que valor toma $(1-\delta_{k\Gamma})\,|a_k\rangle + \delta_{k\Gamma}\,|\gamma_k^\Gamma\rangle$:

- Si $k \in \Gamma \Rightarrow \delta_{k\Gamma} = 1$ y $\gamma_k^\Gamma = \gamma_k$, por lo tanto $(1-\delta_{k\Gamma})\,|a_k\rangle + \delta_{k\Gamma}\,|\gamma_k^\Gamma\rangle = (1-1)\,|a_k\rangle + |\gamma_k\rangle = |\gamma_k\rangle$.
- Si $k \notin \Gamma \Rightarrow \delta_{k\Gamma} = 0$ y $\gamma_k^\Gamma = 0$, por lo tanto $(1-\delta_{k\Gamma})\,|a_k\rangle + \delta_{k\Gamma}\,|\gamma_k^\Gamma\rangle = (1-0)\,|a_k\rangle + 0\,|0\rangle = |a_k\rangle$.

For example, if $\Gamma = \{1,3,7\} \subset \mathbb{N}^8$, $\gamma = (1,0,1) \in \Theta^3$ and $|a\rangle = |0\rangle \in \mathcal{Q}^8$ then

$$|a\rangle_\gamma^\Gamma = |10000010\rangle\,.$$

2.4 Quantum Gates

A single-qubit gate A can be applied over the first qubit of a ket $|a_1a_2\rangle$ and the result is $\mathrm{A}\,|a_1\rangle \otimes |a_2\rangle$. It can also be applied over the last of both qubits and then the result is $|a_1\rangle \otimes \mathrm{A}\,|a_2\rangle$.

Notation 2.5. Let A be a single-qubit gate and $|a\rangle = |a_1 \cdots a_n\rangle$ is a ket with n qubits, the following equivalent notation is used:

$$\bigotimes_{k=1}^{n} \mathrm{A}^{\delta_{hk}}\,|a_k\rangle = \begin{cases} |a_1 \cdots a_h \mathrm{A} a_{h+1} \cdots a_n\rangle \\ |a\rangle\,\mathrm{A}_h \\ |a\rangle_{\mathrm{A}}^{h} \end{cases},$$

where $h \in \mathbb{N}_n$.

Remark 2.6. Throughout this work, gates are written in *postfix* form: the expression $|a\rangle\,\mathrm{A}_h$ denotes the application of the operator A_h to the state $|a\rangle$. Consequently, in a product of gates the leftmost factor is applied first, so expressions are read from left to right in the same order as the corresponding circuit. This convention is purely notational: $|a\rangle\,\mathrm{A}_h\mathrm{B}_l$ stands for $\mathrm{B}_l(\mathrm{A}_h\,|a\rangle)$ in the standard operator notation.

For example, if $|a\rangle = |a_1a_2\rangle$ then

$$\mathrm{A}\,|a_1\rangle \otimes |a_2\rangle = \begin{cases} |a_1 \mathrm{A} a_2\rangle \\ |a\rangle\,\mathrm{A}_1 \\ |a\rangle_{\mathrm{A}}^{1} \end{cases}.$$

Proposition 2.7. Let A be one single-qubit gate. Then, for any two distinct indexes $h, k \in \mathbb{N}_n$

$$\mathrm{A}_k\mathrm{A}_h = \mathrm{A}_h\mathrm{A}_k. \tag{8}$$

Proof. Let $|a\rangle$ be a ket. For any two distinct indexes $h, k \in \mathbb{N}_n$

$$|a\rangle\,\mathrm{A}_k\mathrm{A}_h = |a\rangle_{\mathrm{A}}^{k}\,\mathrm{A}_h = |a\rangle_{\mathrm{AA}}^{kh} = |a\rangle_{\mathrm{A}}^{h}\,\mathrm{A}_k = |a\rangle\,\mathrm{A}_h\mathrm{A}_k.$$

■

By the equality (8) the following notation is well defined.

Notation 2.8. Let $\Gamma = \{k_1, \ldots, k_l\} \subseteq \mathbb{N}_n$ be a set. Let A be a single-qubit gate, the following notation is used:

$$\mathrm{A}_\Gamma = \mathrm{A}_{k_1,\ldots,k_l} = \prod_{k\in\Gamma} \mathrm{A}_k \tag{9}$$

Note that, according to the previous notation, for the specific case of the empty set $\Gamma = \emptyset$, the resulting quantum gate is the identity gate $\mathrm{A}_\emptyset = \mathrm{I}$.

For example, if $|a\rangle = |a_1 a_2\rangle$ and $\Gamma = \{1, 2\}$, then

$$\mathrm{A}\,|a_1\rangle \otimes \mathrm{A}\,|a_2\rangle = \begin{cases} |a_1 \mathrm{A} a_2 \mathrm{A}\rangle \\ |a\rangle\,\mathrm{A}_{1,2} = |a\rangle\,\mathrm{A}_\Gamma \\ |a\rangle_{\mathrm{A}}^{1,2} = |a\rangle_{\mathrm{A}}^{\Gamma} \end{cases} .$$

2.5 More Quantum Gates

Two single-qubit gates A and B can be applied over the first and second qubit respectivement of a ket $|a_1 a_2\rangle$ and the result is $\mathrm{A}\,|a_1\rangle \otimes \mathrm{B}\,|a_2\rangle$.

Notation 2.9. Let A be a single-qubit gate and $|a\rangle = |a_1 \cdots a_n\rangle$ is a ket with n qubits, the following equivalent notation is used:

$$\bigotimes_{k=1}^{n} \mathrm{A}^{\delta_{hk}} \mathrm{B}^{\delta_{lk}}\,|a_k\rangle = \begin{cases} |a_1 \cdots a_h \mathrm{A} a_{h+1} \cdots a_l \mathrm{B} a_{l+1} \cdots a_n\rangle \\ |a\rangle\,\mathrm{A}_h \mathrm{B}_l \\ |a\rangle_{\mathrm{AB}}^{h,l} \end{cases} ,$$

where $h, l \in \mathbb{N}_n$.

For example, if $|a\rangle = |a_1 a_2\rangle$ then

$$\mathrm{A}\,|a_1\rangle \otimes \mathrm{B}\,|a_2\rangle = \begin{cases} |a_1 \mathrm{A} a_2 \mathrm{B}\rangle \\ |a\rangle\,\mathrm{A}_1 \mathrm{B}_2 \\ |a\rangle_{\mathrm{AB}}^{1,2} \end{cases} .$$

Proposition 2.10. Let A and B be two single-qubit gates. Then, for any two distinct indexes $h, k \in \mathbb{N}_n$

$$\mathrm{A}_k \mathrm{B}_h = \mathrm{B}_h \mathrm{A}_k. \tag{10}$$

Proof. Let $|a\rangle$ be a ket. For any two distinct indexes $h, k \in \mathbb{N}_n$

$$|a\rangle\,\mathrm{A}_k \mathrm{B}_h = |a\rangle_{\mathrm{A}}^{k}\,\mathrm{B}_h = |a\rangle_{\mathrm{AB}}^{kh} = |a\rangle_{\mathrm{B}}^{h}\,\mathrm{A}_k = |a\rangle\,\mathrm{B}_h \mathrm{A}_k.$$

■

2.6 Controlled Quantum Gates

Controlled quantum gates are applied to kets formed by multiple qubits so that one or more qubits serve as controllers for a specific operation on one or more qubits of the ket.

The simplest controlled gate A over a ket $|a\rangle = |a_1 \dots a_n\rangle$ has only one control qubit $|a_h\rangle$ and only one target qubit $|a_k\rangle$.

Notation 2.11. Let $|a\rangle = |a_1 \cdots a_n\rangle$ be a ket. Let A be a single-qubit quantum gate. The controlled quantum gate with control qubit $|a_h\rangle$ and target qubit $|a_k\rangle$ is denoted by A_k^h:

$$\begin{aligned} |a\rangle \, \mathrm{A}_k^h &= a_{h0} \, |a\rangle_0^h + a_{h1} \, |a\rangle_1^h \, \mathrm{A}_k = \\ &= a_{h0} \, |a\rangle_0^h + a_{h1} \, |a\rangle_{1\mathrm{A}}^{hk} . \end{aligned} \tag{11}$$

Podemos expresar tambien la puerta cuántica controlada A_k^h como

$$\begin{aligned} |a\rangle \, \mathrm{A}_k^h &= a_{h0} \, |a\rangle_0^h + a_{h1} \, |a\rangle_{1\mathrm{A}}^{hk} = \\ &= a_{h0} \, |a\rangle_0^h + a_{h1} \, |a\rangle_{1\mathrm{A}}^{hk} + a_{h1} \, |a\rangle_1^h - a_{h1} \, |a\rangle_1^h = \\ &= |a\rangle - a_{h1} \, |a\rangle_1^h + a_{h1} \, |a\rangle_{1\mathrm{A}}^{hk} \end{aligned}$$

Proposition 2.12. Let A be a single-qubit gate. Then, for any three distinct indexes $h, k, k' \in \mathbb{N}_n$

$$\mathrm{A}_k^h \mathrm{A}_{k'}^h = \mathrm{A}_{k'}^h \mathrm{A}_k^h \tag{12}$$

Proof. Let $|a\rangle$ be a ket. For any three distinct indexes $h, k, k' \in \mathbb{N}_n$:

$$\begin{aligned} |a\rangle \, \mathrm{A}_k^h \mathrm{A}_{k'}^h &= a_{h0} \, |a\rangle_0^h \, \mathrm{A}_{k'}^h + a_{h1} \, |a\rangle_{1\mathrm{A}}^{hk} \, \mathrm{A}_{k'}^h = \\ &= a_{h0} \, |a\rangle_0^h + a_{h1} \, |a\rangle_{1\mathrm{AA}}^{hkk'} = \\ &= a_{h0} \, |a\rangle_0^h \, \mathrm{A}_k^h + a_{h1} \, |a\rangle_{1\mathrm{A}}^{hk'} \, \mathrm{A}_k^h = \\ &= |a\rangle \, \mathrm{A}_{k'}^h \mathrm{A}_k^h \end{aligned} \tag{13}$$

■

By the proof of (13) the following notation is well defined.

Notation 2.13. Let $|a\rangle = |a_1 \cdots a_n\rangle$ be a ket. Let A be a single-qubit quantum gate. The controlled quantum gate with various control qubits $\Gamma = \{h_1, \dots, h_l\} \subset \mathbb{N}_n$ and various target qubits $\Lambda = \{k_1, \dots, k_m\} \subset \mathbb{N}_n$ such that $\Gamma \cap \Lambda = \emptyset$ is denoted by:

$$\begin{aligned} |a\rangle \, \mathrm{A}_\Lambda^\Gamma = |a\rangle \, \mathrm{A}_{k_1,\dots,k_m}^{h_1,\dots,h_l} &= a_{h_1 0} \, |a\rangle_0^{h_1} + a_{h_1 1} a_{h_2 0} \, |a\rangle_{10}^{h_1 h_2} + \cdots + \\ &+ a_{h_1 1} a_{h_2 1} \cdots a_{h_l 0} \, |a\rangle_{11\dots10}^{h_1 h_2 \dots h_{l-1} h_l} + a_{h_1 1} a_{h_2 1} \cdots a_{h_l 1} \, |a\rangle_{11\dots1\mathrm{A}\dots\mathrm{A}}^{h_1 h_2 \dots h_l k_1 \dots k_m} = \\ &= |a\rangle - a_{h_1 1} a_{h_2 1} \cdots a_{h_l 1} \, |a\rangle_1^\Gamma + a_{h_1 1} a_{h_2 1} \cdots a_{h_l 1} \, |a\rangle_{1\mathrm{A}}^{\Gamma\Lambda} . \end{aligned}$$

Note that, according to the previous notation, for the specific case of the empty set $\Gamma = \emptyset$, the resulting quantum gate is $\mathrm{A}_\Lambda^\Gamma = \mathrm{A}_\Lambda$.

3 General Basic Properties

Some generic properties of gates in a quantum circuit are proven below.

Proposition 3.1. Let A be a single-qubit gate such that $\mathrm{AA} = \mathrm{I}$. Then $\forall h, k \in \mathbb{N}_n$

$$\mathrm{A}_k \mathrm{A}_k = \mathrm{I} \tag{14}$$

$$\mathrm{A}_k^h \mathrm{A}_k^h = \mathrm{I} \tag{15}$$

.

Proof. Let $|a\rangle$ be a ket and two indexes $h, k \in \mathbb{N}_n$. To prove expression (14):

$$|a\rangle \, \mathrm{A}_k \mathrm{A}_k = |a\rangle_{\mathrm{A}}^k \, \mathrm{A}_k = |a\rangle_{\mathrm{AA}}^k = |a\rangle_{\mathrm{I}}^k = |a\rangle \, .$$

Analogously, to prove expression (15), according to (11):

$$\begin{aligned} |a\rangle \, \mathrm{A}_k^h \mathrm{A}_k^h &= a_{h0} \, |a\rangle_0^h \, \mathrm{A}_k^h + a_{h1} \, |a\rangle_{1A}^{hk} \, \mathrm{A}_k^h = \\ &= a_{h0} \, |a\rangle_0^h + a_{h1} \, |a\rangle_{1AA}^{hk} = \\ &= a_{h0} \, |a\rangle_0^h + a_{h1} \, |a\rangle_1^h = |a\rangle \, . \end{aligned}$$

■

Proposition 3.2. Let A and B be two single-qubit gates, and $\Gamma, \Lambda \subset \mathbb{N}_n$ be two sets such that $\Gamma \cap \Lambda = \emptyset$. Then

$$\mathrm{A}_\Gamma \mathrm{B}_\Lambda = \mathrm{B}_\Lambda \mathrm{A}_\Gamma \tag{16}$$

Proof. Let $\Gamma = \{h_1, \ldots, h_l\}$ and $\Lambda = \{k_1, \ldots, k_m\}$ be two subsets of $\mathbb{N}_n$. Then

$$\mathrm{A}_\Gamma \mathrm{B}_\Lambda = \mathrm{A}_\Gamma \mathrm{B}_{k_1} \cdots \mathrm{B}_{k_m} = \mathrm{A}_{h_1} \cdots \mathrm{A}_{h_l} \mathrm{B}_{k_1} \cdots \mathrm{B}_{k_m} .$$

Since $\Gamma \cap \Lambda = \emptyset$, then $h_l \neq k_1$. Thus, according to expression (13), the following is true $\mathrm{A}_{h_l} \mathrm{B}_{k_1} = \mathrm{B}_{k_1} \mathrm{A}_{h_l}$. The same reasoning is valid for all indexes of Γ, so $\mathrm{A}_\Gamma \mathrm{B}_{k_1} = \mathrm{B}_{k_1} \mathrm{A}_\Gamma$. Consequently,

$$\begin{aligned} \mathrm{A}_\Gamma \mathrm{B}_\Lambda = \mathrm{A}_\Gamma \mathrm{B}_{k_1} \cdots \mathrm{B}_{k_m} &= \mathrm{B}_{k_1} \mathrm{A}_\Gamma \mathrm{B}_{k_2} \cdots \mathrm{B}_{k_m} = \\ &= \cdots = \mathrm{B}_{k_1} \cdots \mathrm{B}_{k_m} \mathrm{A}_\Gamma = \mathrm{B}_\Lambda \mathrm{A}_\Gamma \end{aligned}$$

■

Proposition 3.3. Let A be a single-qubit gate. Then, for any two sets $\Gamma, \Lambda \subset \mathbb{N}_n$

$$\mathrm{A}_\Gamma \mathrm{A}_\Lambda = \mathrm{A}_{\Gamma \cup \Lambda} \mathrm{A}_{\Gamma \cap \Lambda} \tag{17}$$

Proof. From basic set theory, $\Gamma \cup \Lambda$ can be expressed as $(\Gamma - (\Gamma \cap \Lambda)) \cup (\Lambda - (\Gamma \cap \Lambda)) \cup (\Gamma \cap \Lambda)$. Then, the following is true

$$\begin{aligned}
\mathrm{A}_\Gamma \mathrm{A}_\Lambda &= \prod_{h\in\Gamma} \mathrm{A}_h \prod_{k\in\Lambda} \mathrm{A}_k = \\
&= \prod_{h\in\Gamma-(\Gamma\cap\Lambda)} \mathrm{A}_h \prod_{h'\in\Gamma\cap\Lambda} \mathrm{A}_{h'} \prod_{k\in\Lambda-(\Gamma\cap\Lambda)} \mathrm{A}_k \prod_{k'\in\Gamma\cap\Lambda} \mathrm{A}_{k'} = \\
&= \prod_{h\in(\Gamma-(\Gamma\cap\Lambda))\cup(\Lambda-(\Gamma\cap\Lambda))\cup(\Gamma\cap\Lambda)} \mathrm{A}_h \prod_{k'\in\Gamma\cap\Lambda} \mathrm{A}_{k'} = \\
&= \prod_{h\in\Gamma\cup\Lambda} \mathrm{A}_h \prod_{k'\in\Gamma\cap\Lambda} \mathrm{A}_{k'} = \mathrm{A}_{\Gamma\cup\Lambda}\mathrm{A}_{\Gamma\cap\Lambda}
\end{aligned}$$

■

Below a theoretical result shows that if a single-qubit gate is applied over a subset of qubits that are eigenstates of the gate with eigenvalue -1, then the result coincides with that of the result with another subset with the same cardinality parity (see [3]). On the other hand, if this latter subset has a different cardinality parity, then the result coincides with the product of the first result with the eigenvalue.

Proposition 3.4. Let A be a single-qubit gate. Let $|a\rangle$ be a ket and $\Gamma, \Lambda \subseteq \mathbb{N}_n$ such that $\forall k \in \Gamma \cup \Lambda, |a\rangle \mathrm{A}_k = -|a\rangle$. Then

$$|a\rangle \mathrm{A}_\Gamma = (-1)^{|\Lambda|-|\Gamma|} |a\rangle \mathrm{A}_\Lambda. \tag{18}$$

Proof. If $\Gamma = \{k_1, \ldots, k_l\}$ where $l = |\Gamma|$, then

$$\begin{aligned}
|a\rangle \mathrm{A}_\Gamma = |a\rangle \prod_{k\in\Gamma} \mathrm{A}_k = |a\rangle \mathrm{A}_{k_1}\mathrm{A}_{k_2}\ldots\mathrm{A}_{k_l} = \\
= (-1)|a\rangle \mathrm{A}_{k_2}\ldots\mathrm{A}_{k_l} = \cdots = (-1)^{|\Gamma|}|a\rangle .
\end{aligned}$$

Analogously, $|a\rangle \mathrm{A}_\Lambda = (-1)^{|\Lambda|}|a\rangle$. Then

$$\begin{aligned}
|a\rangle \mathrm{A}_\Gamma = (-1)^{|\Gamma|}|a\rangle = (-1)^{|\Gamma|-|\Lambda|}(-1)^{|\Lambda|}|a\rangle = \\
= (-1)^{|\Gamma|-|\Lambda|}|a\rangle \mathrm{A}_\Lambda.
\end{aligned}$$

■

4 Standard 1-qubit gates

4.1 Pauli Gates

The well-known Pauli gates $\mathrm{I}, \mathrm{X}, \mathrm{Y}, \mathrm{Z}$ can be expressed according to the notation introduced above as follows:

$$|a\rangle \, \mathrm{I}_k = a_{k0} \, |a\rangle_0^k + a_{k1} \, |a\rangle_1^k \tag{19}$$

$$|a\rangle \, \mathrm{X}_k = a_{k0} \, |a\rangle_1^k + a_{k1} \, |a\rangle_0^k \tag{20}$$

$$|a\rangle \, \mathrm{Y}_k = i a_{k0} \, |a\rangle_1^k - i a_{k1} \, |a\rangle_0^k \tag{21}$$

$$|a\rangle \, \mathrm{Z}_k = a_{k0} \, |a\rangle_0^k - a_{k1} \, |a\rangle_1^k \tag{22}$$

4.2 Pauli Rotation gates

The three Pauli rotation gates RX, RY and RZ are defined by taking exponentials of the respectively Pauli gate.

For a value $\theta \in [0, 2\pi)$ is

$$|a\rangle \, \mathrm{RX}(\theta)_k = \left(a_{k0} \cos\left(\frac{\theta}{2}\right) - a_{k1} \sin\left(\frac{\theta}{2}\right) i\right) |a\rangle_0^k + \left(a_{k1} \cos\left(\frac{\theta}{2}\right) - a_{k0} \sin\left(\frac{\theta}{2}\right) i\right) |a\rangle_1^k \tag{23}$$

$$|a\rangle \, \mathrm{RY}(\theta)_k = \left(a_{k0} \cos\left(\frac{\theta}{2}\right) - a_{k1} \sin\left(\frac{\theta}{2}\right)\right) |a\rangle_0^k + \left(a_{k1} \cos\left(\frac{\theta}{2}\right) + a_{k0} \sin\left(\frac{\theta}{2}\right)\right) |a\rangle_1^k \tag{24}$$

$$|a\rangle \, \mathrm{RZ}(\theta)_k = a_{k0} \left(\cos\left(\frac{\theta}{2}\right) - \sin\left(\frac{\theta}{2}\right) i\right) |a\rangle_0^k + a_{k1} \left(\cos\left(\frac{\theta}{2}\right) + \sin\left(\frac{\theta}{2}\right) i\right) |a\rangle_1^k \tag{25}$$

Proposition 4.1. Let A be one of the three Pauli Matrix X, Y or Z. Let $k \in \mathbb{N}_n$ and $\theta, \varphi \in [0, 2\pi)$.

$$\mathrm{RA}(\theta)_k \mathrm{RA}(\varphi)_k = \mathrm{RA}(\theta + \varphi)_k \tag{26}$$

4.3 Hadamard Gate

The Hadamard gate is a well-known single-qubit gate that maps the basis states $|0\rangle$ and $|1\rangle$ to $|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$ and $|-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$, respectively.

Thus, the Hadamard gate can be expressed using the introduced notation as

follows:

$$\begin{aligned} |a\rangle \, \mathrm{H}_k &= a_{k0} \, |a\rangle_+^k + a_{k1} \, |a\rangle_-^k = \\ &= \frac{a_{k0} + a_{k1}}{\sqrt{2}} \, |a\rangle_0^k + \frac{a_{k0} - a_{k1}}{\sqrt{2}} \, |a\rangle_1^k \end{aligned} \tag{27}$$

Equivalently, since $a_{k+} = \langle + | a_k \rangle = \frac{a_{k0}+a_{k1}}{\sqrt{2}}$ and $a_{k-} = \langle - | a_k \rangle = \frac{a_{k0}-a_{k1}}{\sqrt{2}}$, the action of the Hadamard gate exchanges both bases:

$$|a\rangle \, \mathrm{H}_k = a_{k+} \, |a\rangle_0^k + a_{k-} \, |a\rangle_1^k$$

A Hadamard similarity transform interchanges X and Z gates,

$$\mathrm{H}_k \mathrm{X}_k \mathrm{H}_k = \mathrm{Z}_k \qquad \mathrm{H}_k \mathrm{Y}_k \mathrm{H}_k = \text{-}\mathrm{Y}_k \qquad \mathrm{H}_k \mathrm{Z}_k \mathrm{H}_k = X_k \tag{28}$$

5 Specific Properties of the Controlled Pauli-X Gate

In this section, some specific properties involving the controlled Pauli-X gate are demonstrated.

Proposition 5.1. Let $h, k \in \mathbb{N}_n$ be two indexes. Then

$$\mathrm{X}_k^h \mathrm{Z}_k \mathrm{X}_k^h = \mathrm{Z}_{h,k} \tag{29}$$

Proof. If $|a\rangle$ is a ket and $h \neq k \in \mathbb{N}_n$ are two indexes, then

$$\begin{aligned} |a\rangle \, \mathrm{X}_k^h \mathrm{Z}_k \mathrm{X}_k^h &= a_{h0} \, |a\rangle_0^h \, \mathrm{Z}_k \mathrm{X}_k^h + a_{h1} \, |a\rangle_{1X}^{hk} \, \mathrm{Z}_k \mathrm{X}_k^h = \\ &= a_{h0} \, |a\rangle_{0Z}^{hk} \, \mathrm{X}_k^h + a_{h1} \, |a\rangle_{1ZX}^{hk} \, \mathrm{X}_k^h = \\ &= a_{h0} \, |a\rangle_{0Z}^{hk} + a_{h1} \, |a\rangle_{1XZX}^{hk} = \\ &= a_{h0} \, |a\rangle_{0Z}^{hk} - a_{h1} \, |a\rangle_{1Z}^{hk} = \\ &= a_{h0} \, |a\rangle_0^h \, \mathrm{Z}_k - a_{h1} \, |a\rangle_1^h \, \mathrm{Z}_k = \\ &= |a\rangle \, \mathrm{Z}_{h,k} \end{aligned}$$

■

For the following result it is necessary to introduce the SWAP gate, which acts over two qubits, exchanging the state of both positions:

$$\begin{aligned} |a\rangle \, (SWAP)_{h,k} = a_{h0} a_{k0} \, |a\rangle_{00}^{hk} + a_{h0} a_{k1} \, |a\rangle_{10}^{hk} + \\ + a_{h1} a_{k0} \, |a\rangle_{01}^{hk} + a_{h1} a_{k1} \, |a\rangle_{11}^{hk} \end{aligned} \tag{30}$$

Proposition 5.2. Let $h, k \in \mathbb{N}_n$ be two indexes. Then

$$\mathrm{X}_k^h \mathrm{X}_h^k \mathrm{X}_k^h = (\mathrm{SWAP})_{h,k} \tag{31}$$

Proof. If $|a\rangle$ is a ket and $h \neq k \in \mathbb{N}_n$ are two indexes, then

$$\begin{aligned}
|a\rangle \mathrm{X}_k^h \mathrm{X}_h^k \mathrm{X}_k^h &= a_{h0} |a\rangle_0^h \mathrm{X}_k^h \mathrm{X}_h^k + a_{h1} |a\rangle_{1X}^{hk} \mathrm{X}_k^h \mathrm{X}_h^k = \\
&= a_{h0} |a\rangle_0^h \mathrm{X}_h^k \mathrm{X}_k^h + a_{h1} a_{k0} |a\rangle_{11}^{hk} \mathrm{X}_h^k \mathrm{X}_k^h + \\
&+ a_{h1} a_{k1} |a\rangle_{10}^{hk} \mathrm{X}_h^k \mathrm{X}_k^h = a_{h0} a_{k0} |a\rangle_{00}^{hk} \mathrm{X}_k^h + \\
&+ a_{h0} a_{k1} |a\rangle_{11}^{hk} \mathrm{X}_k^h + a_{h1} a_{k0} |a\rangle_{01}^{hk} \mathrm{X}_k^h + \\
&+ a_{h1} a_{k1} |a\rangle_{10}^{hk} \mathrm{X}_k^h = a_{h0} a_{k0} |a\rangle_{00}^{hk} + \\
&+ a_{h0} a_{k1} |a\rangle_{10}^{hk} + a_{h1} a_{k0} |a\rangle_{01}^{hk} + a_{h1} a_{k1} |a\rangle_{11}^{hk} = \\
&= (\mathrm{SWAP})_{h,k}
\end{aligned}$$

■

Proposition 5.3. Let $h, k \in \mathbb{N}_n$ be two indexes. Then

$$\mathrm{H}_{hk} \mathrm{X}_h^k \mathrm{H}_{hk} = \mathrm{X}_k^h \tag{32}$$

Proof. If $|a\rangle$ is a ket and $h \neq k \in \mathbb{N}_n$ are two indexes, then

$$\begin{aligned}
|a\rangle \mathrm{H}_{kh} \mathrm{X}_h^k \mathrm{H}_{kh} &= a_{h0} |a\rangle_+^h \mathrm{H}_k \mathrm{X}_h^k \mathrm{H}_{kh} + a_{h1} |a\rangle_-^h \mathrm{H}_k \mathrm{X}_h^k \mathrm{H}_{kh} = \\
&= a_{h0} a_{k0} |a\rangle_{++}^{hk} \mathrm{X}_h^k \mathrm{H}_{kh} + a_{h0} a_{k1} |a\rangle_{+-}^{hk} \mathrm{X}_h^k \mathrm{H}_{kh} + \\
&+ a_{h1} a_{k0} |a\rangle_{-+}^{hk} \mathrm{X}_h^k \mathrm{H}_{kh} + a_{h1} a_{k1} |a\rangle_{--}^{hk} \mathrm{X}_h^k \mathrm{H}_{kh} = \\
&= a_{h0} a_{k0} |a\rangle_{++}^{hk} \mathrm{H}_{hk} + a_{h0} a_{k1} |a\rangle_{+-}^{hk} \mathrm{H}_{hk} + \\
&+ a_{h1} a_{k0} |a\rangle_{--}^{hk} \mathrm{H}_{hk} + a_{h1} a_{k1} |a\rangle_{-+}^{hk} \mathrm{H}_{hk} = \\
&= a_{h0} a_{k0} |a\rangle_{+0}^{hk} \mathrm{H}_h + a_{h0} a_{k1} |a\rangle_{+1}^{hk} \mathrm{H}_h + \\
&+ a_{h1} a_{k0} |a\rangle_{-1}^{hk} \mathrm{H}_h + a_{h1} a_{k1} |a\rangle_{-0}^{hk} \mathrm{H}_h = \\
&= a_{h0} |a\rangle_+^h \mathrm{H}_h + a_{h1} |a\rangle_-^h \mathrm{H}_h \mathrm{X}_k = \\
&= a_{h0} |a\rangle_0^h + a_{h1} |a\rangle_1^h \mathrm{X}_k = \\
&= |a\rangle \mathrm{X}_k^h
\end{aligned}$$

■

Proposition 5.4. Let $j, h, k, l \in \mathbb{N}_n$ be four indexes. Then

$$\mathrm{X}_l^{jhk} \mathrm{X}_h \mathrm{X}_l^{jhk} = \mathrm{X}_l^{jk} \mathrm{X}_h \tag{33}$$

Proof. If $|a\rangle$ is a ket and $j, h, k, l \in \mathbb{N}_n$ are four distinct indexes, then the ket $|b\rangle = |a\rangle \, \mathrm{X}_l^{jhk} \mathrm{X}_h \mathrm{X}_l^{jhk}$ is for construction, as shown below

$$
\begin{aligned}
|b\rangle &= a_{j0}a_{h0}a_{k0} \, |a\rangle_{000}^{jhk} \, \mathrm{X}_h \mathrm{X}_l^{jhk} + a_{j0}a_{h0}a_{k1} \, |a\rangle_{001}^{jhk} \, \mathrm{X}_h \mathrm{X}_l^{jhk} + \\
&+ a_{j0}a_{h1}a_{k0} \, |a\rangle_{010}^{jhk} \, \mathrm{X}_h \mathrm{X}_l^{jhk} + a_{j0}a_{h1}a_{k1} \, |a\rangle_{011}^{jhk} \, \mathrm{X}_h \mathrm{X}_l^{jhk} + \\
&+ a_{j1}a_{h0}a_{k0} \, |a\rangle_{100}^{jhk} \, \mathrm{X}_h \mathrm{X}_l^{jhk} + a_{j1}a_{h0}a_{k1} \, |a\rangle_{101}^{jhk} \, \mathrm{X}_h \mathrm{X}_l^{jhk} + \\
&+ a_{j1}a_{h1}a_{k0} \, |a\rangle_{110}^{jhk} \, \mathrm{X}_h \mathrm{X}_l^{jhk} + a_{j1}a_{h1}a_{k1} \, |a\rangle_{111\mathrm{X}}^{jhkl} \, \mathrm{X}_h \mathrm{X}_l^{jhk} = \\
&= a_{j0}a_{h0}a_{k0} \, |a\rangle_{010}^{jhk} \, \mathrm{X}_l^{jhk} + a_{j0}a_{h0}a_{k1} \, |a\rangle_{011}^{jhk} \, \mathrm{X}_l^{jhk} + \\
&+ a_{j0}a_{h1}a_{k0} \, |a\rangle_{000}^{jhk} \, \mathrm{X}_l^{jhk} + a_{j0}a_{h1}a_{k1} \, |a\rangle_{001}^{jhk} \, \mathrm{X}_l^{jhk} + \\
&+ a_{j1}a_{h0}a_{k0} \, |a\rangle_{110}^{jhk} \, \mathrm{X}_l^{jhk} + a_{j1}a_{h0}a_{k1} \, |a\rangle_{111}^{jhk} \, \mathrm{X}_l^{jhk} + \\
&+ a_{j1}a_{h1}a_{k0} \, |a\rangle_{100}^{jhk} \, \mathrm{X}_l^{jhk} + a_{j1}a_{h1}a_{k1} \, |a\rangle_{101\mathrm{X}}^{jhkl} \, \mathrm{X}_l^{jhk} = \\
&= a_{j0}a_{h0}a_{k0} \, |a\rangle_{010}^{jhk} + a_{j0}a_{h0}a_{k1} \, |a\rangle_{011}^{jhk} + \\
&+ a_{j0}a_{h1}a_{k0} \, |a\rangle_{000}^{jhk} + a_{j0}a_{h1}a_{k1} \, |a\rangle_{001}^{jhk} + \\
&+ a_{j1}a_{h0}a_{k0} \, |a\rangle_{110}^{jhk} + a_{j1}a_{h0}a_{k1} \, |a\rangle_{111\mathrm{X}}^{jhkl} + \\
&+ a_{j1}a_{h1}a_{k0} \, |a\rangle_{100}^{jhk} + a_{j1}a_{h1}a_{k1} \, |a\rangle_{101\mathrm{X}}^{jhkl} = \\
&= a_{j0}a_{h0}a_{k0} \, |a\rangle_{000}^{jhk} \, \mathrm{X}_h + a_{j0}a_{h0}a_{k1} \, |a\rangle_{001}^{jhk} \, \mathrm{X}_h + \\
&+ a_{j0}a_{h1}a_{k0} \, |a\rangle_{010}^{jhk} \, \mathrm{X}_h + a_{j0}a_{h1}a_{k1} \, |a\rangle_{011}^{jhk} \, \mathrm{X}_h + \\
&+ a_{j1}a_{h0}a_{k0} \, |a\rangle_{100}^{jhk} \, \mathrm{X}_h + a_{j1}a_{h0}a_{k1} \, |a\rangle_{101\mathrm{X}}^{jhkl} \, \mathrm{X}_h + \\
&+ a_{j1}a_{h1}a_{k0} \, |a\rangle_{110}^{jhk} \, \mathrm{X}_h + a_{j1}a_{h1}a_{k1} \, |a\rangle_{111\mathrm{X}}^{jhkl} \, \mathrm{X}_h = \\
&= a_{j0}a_{k0} \, |a\rangle_{00}^{jk} \, \mathrm{X}_h + a_{j0}a_{k1} \, |a\rangle_{01}^{jk} \, \mathrm{X}_h + \\
&+ a_{j1}a_{k0} \, |a\rangle_{10}^{jk} \, \mathrm{X}_h + a_{j1}a_{k1} \, |a\rangle_{11\mathrm{X}}^{jkl} \, \mathrm{X}_h = \\
&= |a\rangle \, \mathrm{X}_l^{jk} \mathrm{X}_h
\end{aligned}
$$

■

6 Universal Quantum Gates

It can be shown that any quantum gate can be constructed by using a small set of quantum gates. These sets are called a **universal set of quantum gates**.

7 Deutsch Algorithm

The Deutsch Algorithm is one of the first quantum algorithms which showed more efficient than their classical counterpart.

Let $f : \Theta \longrightarrow \Theta$ be a function. There are only four distinct functions, the constant functions 0 and 1, and the balanced functions identity and the opposite or flip function. If we need to know if f is constant or balanced, we need to evaluate f twice, but with a quantum algorithm we only need evaluate once, as was shown by Deutsch [1].

The key observation is that f is constant if and only if $f(0) \oplus f(1) = 0$. The quantum algorithm extracts exactly this single bit of information with one evaluation of f, by querying the oracle on a superposition of both inputs.

Let U be the unitary operator defined on the computational basis by

$$|xy\rangle \, \mathrm{U} = |x(y \oplus f(x))\rangle \, ,$$

where $\oplus$ is the addition mod 2. If we apply the operator U over the $|+-\rangle$ state we have:

$$\begin{aligned} |+-\rangle \, \mathrm{U} &= \frac{1}{\sqrt{2}}(|+0\rangle - |+1\rangle)\mathrm{U} = \frac{1}{2}(|00\rangle + |10\rangle - |01\rangle - |11\rangle)\mathrm{U} = \\ &= \frac{1}{2}(|0(0 \oplus f(0))\rangle + |1(0 \oplus f(1))\rangle - |0(1 \oplus f(0))\rangle - |1(1 \oplus f(1))\rangle) = \\ &= \begin{cases} \frac{1}{2}(|00\rangle + |10\rangle - |01\rangle - |11\rangle) = |+-\rangle & \text{if } f(0) = 0 \text{ and } f(1) = 0 \\ \frac{1}{2}(|00\rangle + |11\rangle - |01\rangle - |10\rangle) = |--\rangle & \text{if } f(0) = 0 \text{ and } f(1) = 1 \\ \frac{1}{2}(|01\rangle + |10\rangle - |00\rangle - |11\rangle) = -|--\rangle & \text{if } f(0) = 1 \text{ and } f(1) = 0 \\ \frac{1}{2}(|01\rangle + |11\rangle - |00\rangle - |10\rangle) = -|+-\rangle & \text{if } f(0) = 1 \text{ and } f(1) = 1 \end{cases} \end{aligned}$$

so, the resulting state is, up to a global phase:

$$|+-\rangle \, \mathrm{U} = (-1)^{f(0)} \begin{cases} |+-\rangle & \text{if } f \text{ is constant} \\ |--\rangle & \text{if } f \text{ is balanced} \end{cases} .$$

Let us consider a quantum circuit which implements the Deutsch algorithm. According to the postfix convention, the circuit is read from left to right:

$$\mathrm{D} = \mathrm{X}_2\mathrm{H}_{1,2}\mathrm{U}\mathrm{H}_1$$

Applying the circuit to the initial state $|00\rangle$ and using the computation above:

$$\begin{aligned} |00\rangle \, \mathrm{D} &= |00\rangle \, \mathrm{X}_2\mathrm{H}_{1,2}\mathrm{U}\mathrm{H}_1 = |01\rangle \, \mathrm{H}_{1,2}\mathrm{U}\mathrm{H}_1 = |+-\rangle \, \mathrm{U}\mathrm{H}_1 = \\ &= (-1)^{f(0)} \begin{cases} |+-\rangle \, \mathrm{H}_1 \\ |--\rangle \, \mathrm{H}_1 \end{cases} = (-1)^{f(0)} \begin{cases} |0-\rangle & \text{if } f \text{ is constant} \\ |1-\rangle & \text{if } f \text{ is balanced} \end{cases} \end{aligned} \tag{34}$$

Therefore, measuring the first qubit in the computational basis yields 0 if f is constant and 1 if f is balanced, with a single evaluation of U.

References

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